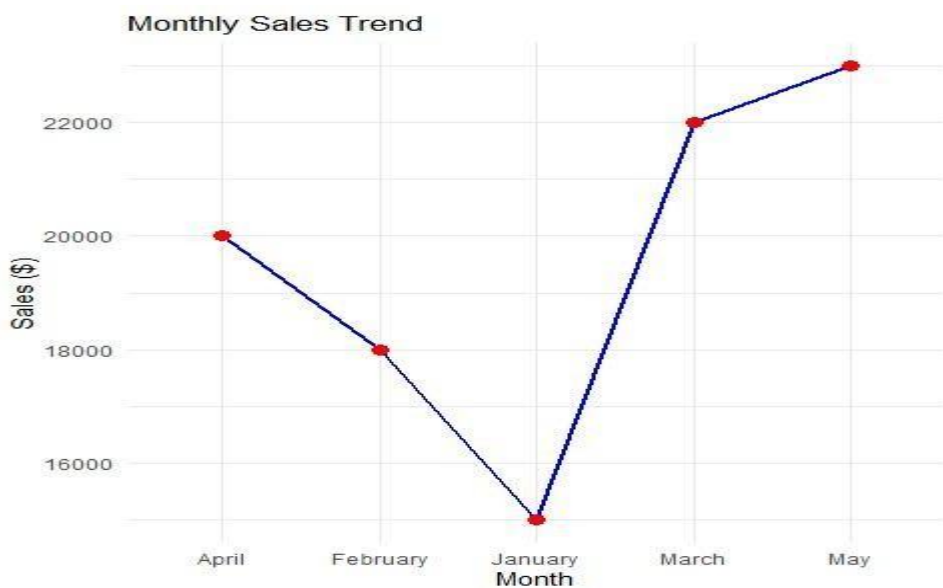


SET 18- Time Series Analysis.

18 A. Time Series Line Chart

Program:

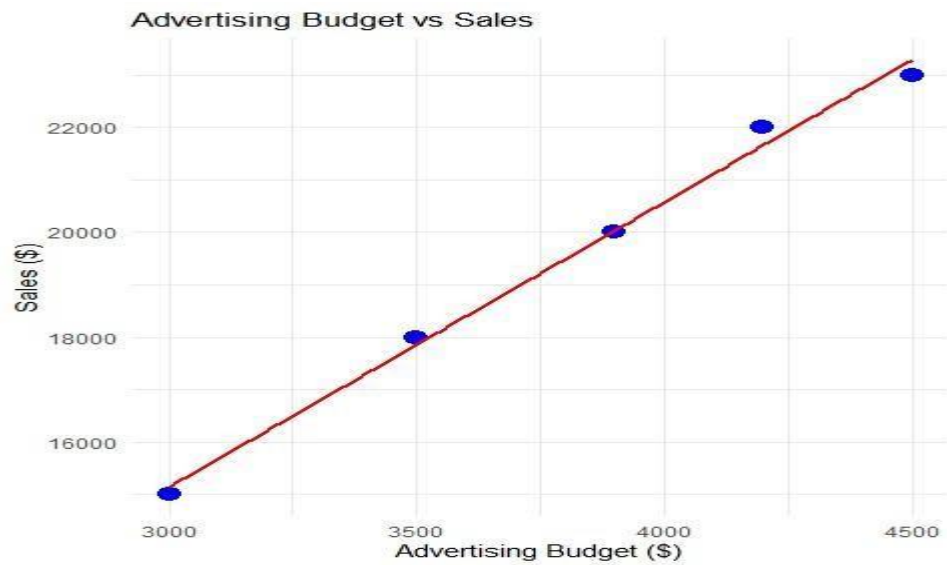
```
sales_data <- data.frame(  
  Month=c("January","February","March","April"  
  ,"May"),  
  Sales=c(15000,18000,22000,20000,23000))  
ggplot(sales_data, aes(x=Month, y=Sales,  
  group=1))+ geom_line(color="blue",linewidth=1)+  
  geom_point(color="red",size=3)+  
  labs(title="Monthly Sales Trend", x="Month",  
  y="Sales ($)")+ theme_minimal() Output:
```



18 B. Scatter Plot

Program:

```
sales_data$Advertising_Budget <- c(3000,3500,4200,3900,4500) ggplot(sales_data,  
  aes(x=Advertising_Budget, y=Sales))+ geom_point(size=4,color="blue")+  
  geom_smooth(method="lm",se=FALSE,color="red")+ labs(title="Advertising Budget vs  
  Sales", x="Advertising Budget ($)", y="Sales ($)")+  
  theme_minimal() Output:
```



18 D. Autocorrelation Plot

Program:

```
sales_ts <- ts(
sales_data$Sales,
frequency=12)
acf(
sales_ts, main="Autocorrelation Plot of
```

Monthly Sales")

Output:

